



UITs

Future will be better than the past

University of Information Technology & Sciences

An initiative of *PHP Family*

Microprocessors, Microcontrollers and Assembly Language Lab

Lab Report - 5

Course Title : Microprocessors, Microcontrollers and Assembly
Language Lab

Course Code : CSE0611324

Submitted to:

Teacher's Name	Designation
Tahmina Yeasmin Lima	Lecturer
Abrar Mahbub Tanim	Lecturer

Submitted by:

Student Name : Gaus Saraf Murady

Student ID : 0432410005101088

Semester : Autumn 2026

Batch : 55

Department : CSE

Date of Submission: 10.09.26

Experiment No : 5

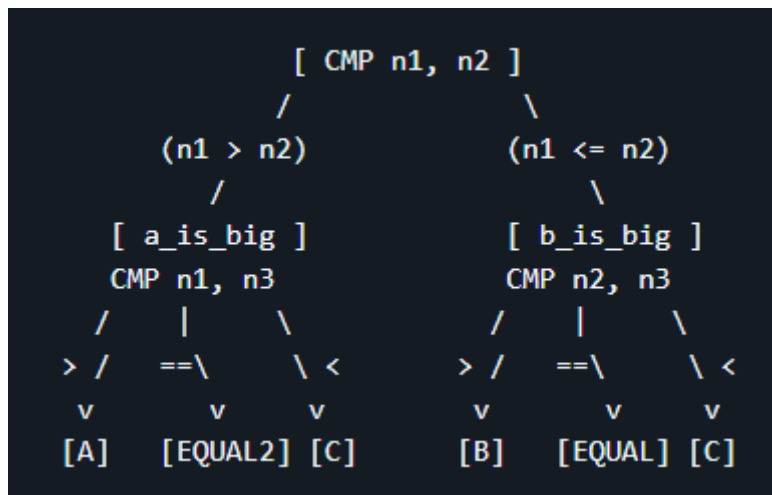
Experiment Name: Find the biggest number among 3 numbers

Introduction:

In this lab, we worked with comparing numbers to find the largest among them. In the lab, we worked with 2 numbers and for the assignment, we were tasked with finding the largest among 3 numbers.

For my report, I have implemented only the “largest among 3 numbers” case, with a noticeable edge case of equal comparisons. I have addressed this by concluding that the **first layer comparison check** implies that I can jump to any of the 2 subsequent comparisons to make a decision on the final number’s state (largest / equal). Hence, I chose **je b_is_big** (I could choose **je a_is_big** as well; resulting in the same)

Implemented code:



Code:

```
.model small
.stack 100h

.data          ;data segment
    a_input db 'Enter the value of A: $'
    b_input db 10, 13, 'Enter the value
of B: $'
    c_input db 10, 13, 'Enter the value
of C: $'

    a db 10, 13, ' A is largest!$'
    b db 10, 13, ' B is largest!$'
    c db 10, 13, ' C is largest!$'

    equal db 10, 13, ' Equal values!$'

    n1 db ?
    n2 db ?
    n3 db ?

.code
    main proc
        mov ax,@data ;move the data
segment in ax
        mov ds, ax

        mov ah, 9
        lea dx, a_input
        int 21h
        mov ah, 1
        int 21h
        mov n1, al

        mov ah, 9
        lea dx, b_input
        int 21h
        mov ah, 1
        int 21h
        mov n2, al
```

```
        mov ah, 9
        lea dx, c_input
        int 21h
        mov ah, 1
        int 21h
        mov n3, al

        ;cmp bl, bh ; compare
        mov al, n1
        mov ah, n2
        cmp al, ah
        jg a_is_big ; if bl bigger
than bh then jg flag on
        jl b_is_big ; if bh bigger
than bl then jl flag on
        je b_is_big ; in case of equal
case we can compare any one with the 3rd

a_is_big:
        ; bl is big so print variable a

        mov al, n1
        mov ah, n3

        cmp al, ah
        jg a_is_big2
        jl c_is_big
        je both_are_equal2

b_is_big:

        mov al, n2
        mov ah, n3

        cmp al, ah
        jg b_is_big2
        jl c_is_big
        je both_are_equal

b_is_big2:
        mov ah, 9
        lea dx, b
        int 21h
        jmp exit
```

```
both_are_equal:

    mov ah, 9
    lea dx, equal
    int 21h
    jmp exit
```

```
a_is_big2:
    mov ah, 9
    lea dx, a
    int 21h
    jmp exit
```

```
c_is_big:
    mov ah, 9
    lea dx, c
    int 21h
```

```
    jmp exit

both_are_equal2:
    mov ah, 9
    lea dx, equal
    int 21h

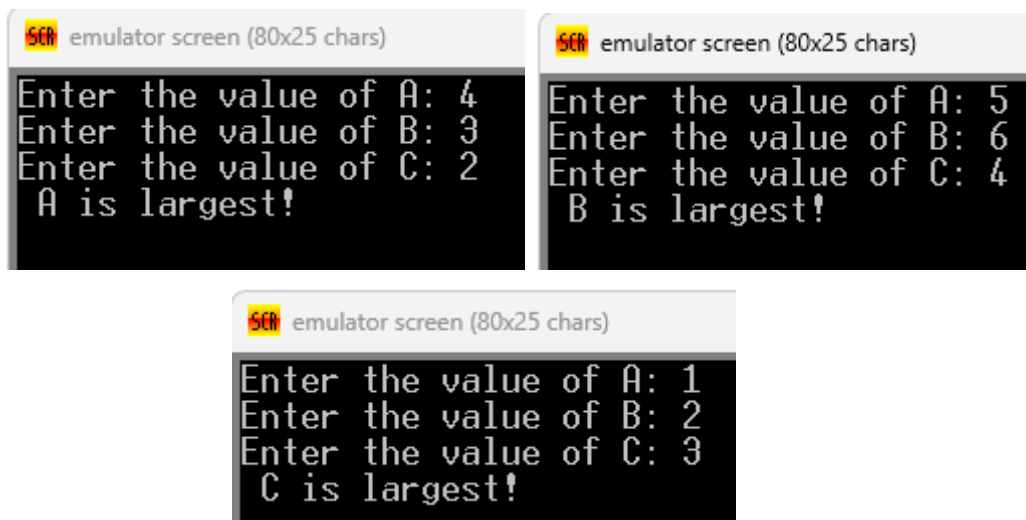
    jmp exit

exit:

    mov ah, 4ch

main endp
end main
```

Output:



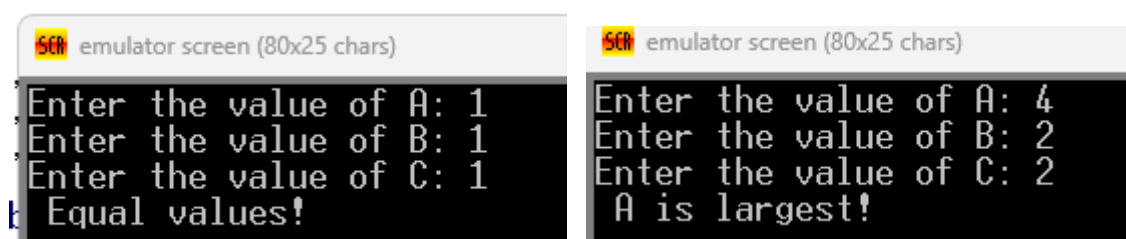
The first screenshot shows the output for inputs A=4, B=3, C=2, where A is the largest. The second screenshot shows the output for inputs A=5, B=6, C=4, where B is the largest. The third screenshot shows the output for inputs A=1, B=2, C=3, where C is the largest.

```
emulator screen (80x25 chars)
Enter the value of A: 4
Enter the value of B: 3
Enter the value of C: 2
A is largest!

emulator screen (80x25 chars)
Enter the value of A: 5
Enter the value of B: 6
Enter the value of C: 4
B is largest!

emulator screen (80x25 chars)
Enter the value of A: 1
Enter the value of B: 2
Enter the value of C: 3
C is largest!
```

(Equal Cases)



The first screenshot shows the output for inputs A=1, B=1, C=1, where all values are equal. The second screenshot shows the output for inputs A=4, B=2, C=2, where A is the largest.

```
emulator screen (80x25 chars)
Enter the value of A: 1
Enter the value of B: 1
Enter the value of C: 1
Equal values!

emulator screen (80x25 chars)
Enter the value of A: 4
Enter the value of B: 2
Enter the value of C: 2
A is largest!
```

Discussions:

The relevant notes have been documented at:

- [Microprocessor_Assembly_Lab/LabCodes/Learning/9.1-theory.md at main · b1tranger/Microprocessor_Assembly_Lab](#)
- [Microprocessor_Assembly_Lab/LabCodes/Learning/9.2_assignment-theory.md at main · b1tranger/Microprocessor_Assembly_Lab](#)
- [Microprocessor_Assembly_Lab/LabCodes/Learning/9.3_alt_assignment-theory.md at main · b1tranger/Microprocessor_Assembly_Lab](#)

Conclusion:

I have successfully completed the assignment. Although I have addressed the **Equal Numbers Case**, and I have sufficiently addressed the requirements for this assignment.